



Fox School of Business

TEMPLE UNIVERSITY®

Department of Risk, Actuarial, Healthcare Management, Legal Studies
AS 3504/5108 Actuarial Analytics

Analytics Practice Exam #1 Solutions

Student Name: _____

Score: _____ / _____

75 Minutes. 40 Points.

Show all your work on this exam.

The more I understand your thought process, the more partial credit I can give, but writing a bunch of formulas down on a Q you have no idea how to start is not work.

No work = No credit for numerical questions.



Conceptual Questions Section (1 Point Each x 15). Clearly circle the True statements.

1. In simple linear regression with 1 predictor, the standard error of the regression is equivalent to the variance of the response for a given x .
2. Unsupervised problems are problems without a clear predictor.
3. If a scatterplot shows all the points forming a straight line with a slope of 0.5, the variables' sample correlation is near 0.5.
4. The goal of modeling should be to reduce the variability of the error term ϵ in a supervised learning model.
5. In simple linear regression, both the confidence interval and prediction interval should be narrowest at the sample mean of Y .
6. Categorical variables can only take on one of a finite number of different values.
7. In K-Nearest Neighbors, as k increases, flexibility and variance both decrease.
8. Under hypothesis testing, a large p -value corresponds with a smaller $|t|$ -statistic.
9. The Bayes' error rate for a categorical response cannot be over 50%.
10. Predicting whether a transaction is fraudulent or not fraudulent is a classification problem, even if the model internally estimates a probability of fraud.
11. Linear regression is non-parametric because it assumes a specific functional form with free parameters to estimate.
12. If a model estimates the true function perfectly, the test MSE should be below the irreducible error.
13. The best predictive model typically has the lowest variance and lowest squared bias among all candidate models.
14. In simple linear regression, normally distributed independent errors imply normally distributed independent responses conditional on the predictor values.
15. If the estimated slope in simple linear regression is -9, it is interpreted as each one-unit increase in x makes the predicted y decrease by -9.



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Conceptual Questions Solutions:

1. False; standard error squared estimates
2. False; clear response
3. False; If everything falls on a line w/positive slope, the correlation is 1.
4. False; cannot reduce
5. False; X
6. True
7. True
8. True
9. False; 3 categories w/equal probability as counter-example
10. True
11. False; parametric
12. False; irreducible error cannot be eliminated
13. False; bias-variance trade-off
14. True
15. False; decrease by -9 means increase by 9



Computation Questions Section (5 Points Each x 5)

16. Consider the following dataset:

y_i (Favored Team)	x_i (Distance from Philadelphia)
Eagles	10
Eagles	35
Eagles	70
Steelers	50
Steelers	100
Steelers	140

- A. (4 points) Using the K-nearest neighbors algorithm with $K = 3$, find the smallest integer distance x_i that would result in a Steelers prediction. Explain your reasoning and work if you find a logical approach. If you can't reason it out, you can attempt trial and error.
- B. (1 point) Would changing K to 5 result in a more or less flexible model?

Sol: Bayes/KNN

Boundary would include the 70, 50, 100 points, with the 100 beating out the 35.

$$x - 35 > 100 - x$$

$$2x > 135$$

$x > 67.5$ or $x = 68$ is the minimum distance that predicts Steelers.

Higher K results in a less flexible model.



17. An investigative journalist examines the relationship between amount of ice cream sold (explanatory) and crime rate (response).

A survey of eight randomly selected cities yielded the following info:

$$\bar{x} = 4$$

$$\bar{y} = 22$$

$$s_x = 1.9$$

$$s_y = 11.3$$

$$\text{Cov}(X, Y) = 12.5$$

- A. (4 points) Calculate the coefficient of determination, rounded to the nearest percent.
B. (1 point) Explain the meaning of the coefficient of determination in the context of the problem.

Sol: SLR

$$R^2 = \frac{12.5^2}{1.9^2 * 11.3^2} = 34\%$$

34% of the variability in crime rate can be explained by the linear relationship between ice cream sold and crime rate.



18. You fit the following model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ to 21 observations to predict the number of fantasy football points scored.

Using the method of least squares, you find $b_1 = 5$ and $se(\hat{\beta}_1) = 2.32$.

- A. (2 points) What is a 2-sided 95% confidence interval for b_1 ?
- B. (2 points) From conducting the hypothesis test $H_0: \beta_1 = 1$ vs. $H_a: \beta_1 \neq 1$, what range must the p -value fall in, based on available Table data?
- C. (1 point) Using typical hypothesis testing criteria, what is the conclusion of your hypothesis test from B?

Sol: SLR Inferences

$$df = 21 - 2 = 19$$

2-sided critical $t = 2.093$ (double the t-table values)

$$5 \pm 2.093(2.32) = (0.14, 9.86)$$

$$t = \frac{5 - 1}{2.32} = 1.724$$

$$1.3277 < 1.724 < 1.7291 \text{ so } 0.2 > p > 0.1$$

We fail to reject the null hypothesis that $b_1 = 1$ even at $\alpha = 10\%$.



19. You fit a simple linear regression model to the dataset.

x_i (# of Children)	y_i (# of Labubu Owned)	$\hat{y}_i$
4	5	5
6	7	6
8	9	7
10	8	8
12	11	9

- A. (3 points) Calculate the mean-squared error and residual standard error.
B. (2 points) Interpret the intercept coefficient in the context of the problem.

Sol: SLR

$$e_i = y_i - \hat{y}_i$$

Residuals are: 0, 1, 2, 0, 2

$$SSE = 0^2 + 1^2 + 2^2 + 0^2 + 2^2 = 9$$

$$MSE = \frac{SSE}{n-2} = \frac{9}{5-2} = 3$$

$$RSE = s = \sqrt{3}$$

Use the x_i and $\hat{y}_i$ to back into the fitted equation: $\hat{y}_i = 3 + 0.5x_i$.

The predicted/expected # of Labubu owned is 3 when the # of children is 0, using the linear model.



20. You are given the following R output for a simple linear regression model:

Call:

```
lm(formula = y ~ x)
```

Coefficients:

	Estimate	Std. Error
(Intercept)	8.000	2.500
x	0.900	0.120

Residual standard error: 3.100 on 48 degrees of freedom

- (4 points) You want to predict the response variable when the explanatory variable is $x = 30$. Calculate the standard error of this prediction.
- (1 point) What do you expect the standard error of the prediction to do as x increases above 30?

Sol: SLR Inferences

$$n = 48 + 2 = 50$$

$$se(b_1) = \sqrt{\frac{MSE}{\sum_i (x_i - \bar{x})^2}}$$

$$0.12 = \frac{3.1}{\sqrt{\sum_i (x_i - \bar{x})^2}}$$

$$\sum_i (x_i - \bar{x})^2 = 667.361$$

$$se(b_0) = \sqrt{MSE \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_i (x_i - \bar{x})^2} \right)}$$

$$2.5 = 3.1 \sqrt{\frac{1}{50} + \frac{\bar{x}^2}{667.361}}$$

$$\bar{x} = 20.51$$

$$se(\hat{y}_{n+1}) = \sqrt{MSE \left(1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right)}$$

$$se(\hat{y}_{n+1}) = 3.1 \sqrt{1 + \frac{1}{50} + \frac{(30 - 20.51)^2}{667.361}} = 3.33$$

As $x > 30$ gets further away from $\bar{x}$, the standard error of the prediction will increase.